Harshita Sharma

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Education

2024 – 2028 B.Tech, **IIT Bhilai, Mechanical Engineering** **CGPA: 8.29**

Experience

AI Engineer Intern – APTransco (State Load Dispatch Centre) *Present*

- Build short-horizon forecasting models (LSTM, Transformer) on high-frequency demand time-series, benchmarked against statistical and gradient-boosted baselines with walk-forward, out-of-sample evaluation.
- Engineer predictive features from noisy multi-source signals – weather, festival calendars, agricultural pump usage, industrial load cycles – and quantify each one's marginal contribution to forecast error.
- Mine statewide consumption data for recurring load patterns and regime shifts, isolating the systematic drivers of demand variability and over/under-procurement.
- Run reproducible Python research workflows on Linux and translate model output into scheduling recommendations for non-technical grid operators.

Tools: Python, Linux, PyTorch, TensorFlow, LSTM, Transformer, Pandas, NumPy, Time-Series Analysis

Skills

Languages: C++, **Python**, C, SQL
Quantitative / ML: Time-Series Forecasting, Statistical Modeling, Feature Engineering, Pattern Recognition, Gradient Boosting (XGBoost, LightGBM, CatBoost), Ensembling & Stacking, Cross-Validation, LSTMs & Transformers, Scikit-learn, PyTorch, TensorFlow
Data & Tooling: Pandas, NumPy, Matplotlib, Seaborn, Jupyter/Colab, Git
Systems: Linux (Ubuntu), Bash, ROS2, Gazebo, PX4, MAVROS, Pixhawk, Raspberry Pi

Projects

Gridlock Hackathon – Traffic Demand Prediction (Flipkart) Notebook

- Built a stacked ensemble of LightGBM, XGBoost, and CatBoost to forecast city-wide traffic demand in Bangalore, reaching an OOF R^2 of **95.43** under a leakage-free cross-validation setup.
- Engineered lag, rolling-window, and cyclical time features from spatiotemporal data; used out-of-fold predictions as meta-features to cut variance across the blend.
- Placed **3rd** in a time-boxed hackathon: framed an open-ended problem, shipped a working solution under deadline pressure, and defended the approach to judges.

Tools: Python, LightGBM, XGBoost, CatBoost, Pandas, NumPy

Asteroid Risk Classification Notebook

- Won **1st position** in an Intra-IIT ML Challenge: classified asteroids as hazardous vs. non-hazardous from orbital and physical features on a heavily imbalanced dataset.
- Compared Random Forest against XGBoost, trading overall accuracy (**82%**) against hazardous-class recall to pick the model with the right error profile for an asymmetric-cost decision.

Tools: Python, Scikit-learn, Random Forest, XGBoost, Pandas, NumPy, Matplotlib

Renewable Energy Forecasting Notebook

- Analyzed solar and wind generation time-series against temperature and wind-speed covariates, extracting seasonality, trend, and regime-shift structure from the signal.
- Ran preprocessing, feature engineering, and correlation analysis to identify which covariates carry genuine predictive signal, and communicated findings through visualizations.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Google Colab

Autonomous Precision Landing of Quadrotor on ArUco Marker View | Repo

- Delivered a real-time perception-to-control pipeline (marker detection → pose estimate → setpoint command) running under latency constraints on embedded hardware.
- Integrated PX4, Gazebo, ROS2 Humble, MAVROS, Pixhawk 2.4.8, and Raspberry Pi 5 on Linux; validated in simulation before deploying to the physical build.

Tools: PX4, ROS2 Humble, Gazebo, MAVROS, OpenCV (ArUco), Pixhawk 2.4.8, Raspberry Pi 5, Python, Linux

Achievements and Leadership

- **Maintainer – OpenLake, IIT Bhilai:** Lead open-source project maintenance, review contributions, and coordinate a distributed team of contributors.
- **Core Member – Blockchain Society, IIT Bhilai:** Organize technical workshops and hackathons end-to-end – logistics, speakers, and participants – while running multiple events in parallel.
- **Third Position – Summer ChainRxn Bootcamp** View
  Designed and built an animated front-end landing page with a team under competition deadlines.
- **MATLAB ML Certifications** Cert 1 | Cert 2
  Machine Learning Onramp and Classification Models.